

Multi-Source Explainable AI for Blockchain Transaction Manipulation Detection Using Supervised and Unsupervised Models

Isaac Kofi Nti

ntiik@ucmail.uc.edu

University of Cincinnati

Owusu Nyarko-Boateng

University of Cincinnati

Samuel Boateng

University of Cincinnati

Benjamin Asubam Weyori

University of Energy and Natural Resources

Adebayo Felix Adekoya

Catholic University College of Ghana

Research Article

Keywords: Explainable Artificial Intelligence (XAI), Blockchain Security, Ethereum Fraud Detection, Machine Learning for Anomaly Detection, XGBoost, LightGBM

Posted Date: June 17th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-6549946/v1>

License:  This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Multi-Source Explainable AI for Blockchain Transaction Manipulation Detection Using Supervised and Unsupervised Models

Isaac Kofi Nti^{1,2}, Owusu Nyarko-Boateng¹, Samuel Boateng¹, Benjamin Asubam Weyori³ and Adebayo Felix Adekoya⁴

¹School of Information Technology, University of Cincinnati, Cincinnati, USA,

²Information Technology and Analytics Center (ITAC), School of Information Technology, University of Cincinnati, Cincinnati, Cincinnati, USA

³Department of Electrical and Computer Engineering, University of Energy and Natural Resources, Sunyani, Ghana

⁴Catholic University of Ghana, Sunyani-Fiapre, Ghana

Abstract

Blockchain has changed the digital finance systems through decentralized, transparent, and tamper-resistant transactions. However, it has led to sophisticated manipulation schemes, like wash trading, rug pulls, and flash loan exploits, due to its open nature. These threats undermine trust and stability in decentralized markets. Machine Learning (ML) models have shown promise in detecting such anomalies. However, ML's black-box nature poses significant challenges for forensic analysis, user accountability, and regulatory compliance. Hence, to identify and interpret manipulated transaction patterns on public blockchain networks, we propose a novel framework that combines Explainable Artificial Intelligence (XAI) with an ML-based fraud detection scheme. This study proposes a multi-source explainable AI framework for detecting manipulated blockchain transactions using engineered features derived from Ethereum transactional data, scam address blacklists, and phishing datasets. We integrated supervised learning models (XGBoost, LightGBM) and unsupervised approaches (autoencoder neural network, isolation forest) to detect anomalies. We apply SHapley Additive exPlanations (SHAP) to quantify the contribution of each feature in real-time anomaly predictions, offering both global and instance-level explanations. Our model achieves high detection accuracy (F1-score > 0.96) while significantly improving transparency in fraud classification. Furthermore, our visual interpretations with SHAP force plots and transaction graphs permit intuitive auditing of suspicious activities. Our proposed fusion framework of detection and interpretability provides a robust foundation for automated blockchain forensics. This framework helps auditors, developers, and regulators understand the rationale behind flagged anomalies. Our proposed framework advances the development of transparent, scalable, and effective fraud detection systems for decentralized environments.

Keywords: Explainable Artificial Intelligence (XAI), Blockchain Security, Ethereum Fraud Detection, Machine Learning for Anomaly Detection, XGBoost, LightGBM

1. Introduction

Blockchain technology has changed the nature of financial transactions, offering decentralization, transparency, and impenetrable systems (Chahar et al., 2025; Chowdhury et al., 2024; Kilic et al., 2022; Pasupuleti, 2025). The technology was originally designed to support digital currencies. However, blockchain's immutable ledgers and consensus mechanisms have since been embraced in diverse sectors. These areas include finance, governance, healthcare, and supply chains, where secure and auditable records are critical (Kovalenko, 2025, and Rane et al., 2023). Despite these advances, the open and pseudonymous nature of blockchain networks has given rise to a new wave of manipulation strategies. This type of fraud includes "wash trading," "pump-and-dump" schemes, and "flash loan" exploits. These schemes pose a threat to the trust and stability of decentralization systems, as noted by Ikeda et al. (2025), Mounnan et al. (2024), and Osterrieder et al. (2024). Additionally, the rise of Decentralized Finance (DeFi) and Non-Fungible Tokens (NFTs) has resulted in the growth of these risks. In these environments, high-frequency trades, minimal regulation, and anonymous actors provide fertile ground for market manipulation and fraud.

In recent years, blockchain ecosystems have witnessed a significant surge in manipulation and fraudulent activity, with 2024 marking one of the most damaging periods to date. According to Chainalysis, crypto hacking incidents rose to 303 attacks in 2024, resulting in an estimated \$2.2 billion in stolen funds—representing a 21% increase from the previous year (Chainalysis Team, 2025; Mondoh, 2025; Mattackal, 2025; Singh, 2024). A substantial portion of these breaches targeted decentralized finance (DeFi) platforms, known for their rapid development cycles and comparatively weaker security protocols. Disturbingly, groups affiliated with North Korea were responsible for \$1.34 billion (61% of the total crypto theft). Apart from hacking, scam activity also reached record highs. For instance, fraudulent crypto schemes generated a minimum of \$9.9 billion in 2024. The total is projected to rise to \$12.4 billion as more data becomes available (Chainalysis Team, 2025; Mondoh, 2025; Mattackal, 2025; Singh, 2024). Also, scams such as "pig butchering," using long-term social engineering and trust-building to defraud

victims, grew by nearly 40%. Pig butchering and high-yield investment scams were reported to account for over 80% of all crypto scam revenue. Despite this rise in targeted manipulation, the overall proportion of illicit crypto transactions fell by 51% to \$45 billion—just 0.4% of total transaction volume. This data suggests that while the overall use of blockchain is increasing, malicious activities are becoming more concentrated and sophisticated (Chainalysis Team, 2025; Mondoh, 2025; Mattackal, 2025; Singh, 2024). These trends underscore the urgent need for advanced anomaly detection systems (Osterrieder et al., 2024) and XAI frameworks to ensure trust and accountability within decentralized systems.

Conventional tools for detecting anomalies in financial systems often underperform when applied to decentralized and pseudonymous networks (Ashfaq et al., 2022; Chitsungo, 2024; Hernandez Aros et al., 2024; Kilic et al., 2022; Nadella et al., 2024). Hence, there is an urgent call for advanced and sophisticated techniques for anomaly detection that provide transparent and interpretable justifications (Hernandez Aros et al., 2024; Kumar et al., 2024; Kshetri, 2025; Svetinovic, 2019). Such an approach is a fundamental necessity for the fields of forensics, compliance, and public trust. Therefore, as pointed out by Bello et al. (2024) and Kayikci & Khoshgoftaar (2024), putting machine learning and blockchain together can create a powerful and fast way to stop fraud, making the whole financial world a safer place. As such, recent work has explored the use of machine learning (ML) and anomaly detection for identifying abnormal behaviors in blockchain transactions (Ikeda et al., 2025; Hasan et al., 2024; Tabaghsar et al., 2024). Despite the improvement in automated ML-based anomaly detection, the “black box” nature of most ML algorithms remains a challenge (Hernandez Aros et al., 2024; Jayanthi et al., 2024; Vajjhala et al., 2025). Their lack of interpretability inhibits adoption in high-stakes domains like regulatory compliance, smart contract auditing, and institutional finance (Kumar et al., 2024; Rane et al., 2023). Moreover, several existing ML models focused on the unique structure and metadata of blockchain transactions, limiting their effectiveness and transparency. Furthermore, most existing works usually use graph-based factors, transaction flow analysis, and temporal patterns to detect malicious activity.

Hence, we propose an ML framework that integrates Explainable AI (XAI) to detect and interpret manipulated transaction patterns in blockchain networks. Specifically, we focus on leveraging SHapley Additive exPlanations (SHAP) to quantify feature contributions in both supervised (XGBoost and LightGBM) and unsupervised (Isolation Forest and Autoencoder) anomaly detection models. Our goal is to identify manipulation with high fidelity and provide human-understandable explanations. This human focus on explainability supports downstream auditing and decision-making processes (Chahar et al., 2025; Ikeda et al., 2025; Kumar et al., 2024). We collected Ethereum-based transactional records and open-source datasets. We constructed a multi-source dataset to support robust fraud detection by combining three distinct data streams. These are (1) raw transactional and token activity scraped from the Ethereum blockchain via the Etherscan API, (2) public scam address blacklists from MyEtherWallet and Etherscan’s tagging system, and (3) pre-labeled phishing and fraud transaction datasets sourced from Kaggle’s Ethereum phishing scams dataset (Eth-PSD). The combination of these sources provides complementary perspectives: live blockchain behavior, verified scam metadata, and human-labeled fraud patterns. The dataset includes features derived from temporal patterns, transaction values, address activity, and contract interactions. We then matched sender, and receiver addresses against known scam lists using label propagation. Next, we implemented a preprocessing pipeline that included feature engineering, normalization, and Synthetic Minority Over-sampling Technique (SMOTE)-based balancing to address the issue of class imbalance. Our study’s main contributions to both the technical and operational dimensions of blockchain security are as follows:

1. We constructed a multi-source labeled dataset that reflects real-world blockchain behavior and manipulated transaction signatures.
2. Proposed a feature engineering strategy that extracts behavioral, temporal, and economic transaction factors significant to fraud detection.
3. Applied model explainability with SHAP to enhance trust and interpretability in fraud detection systems.
4. We evaluated and compared supervised and unsupervised models and showed the significance of label-rich behavioral signals in model accuracy.
5. Offers a reproducible and modular framework that can support auditing, compliance, and forensic analysis in decentralized financial environments.

This paper is organized as follows. Section 2 reviews related work on blockchain anomaly detection and explainable AI. Section 3 presents the proposed architecture, data sources, and model selection. Section 4 describes the XAI integration and visualization techniques. Section 5 evaluates the performance of our approach across various manipulation scenarios. Section 6 concludes with key findings, limitations, and directions for future research.

2. Related Work

2.1 Blockchain Manipulation and Anomaly Detection

The security and transparency of blockchain systems, while foundational to their design, are frequently undermined by manipulation strategies such as wash trading, front-running, and flash loan attacks. As noted by Ikeda et al. (2025), these manipulations exploit the open, decentralized structure of crypto-assets, with significant consequences for public trust and economic stability. The authors introduced a multi-indicator anomaly detection framework. The framework combines topological features (e.g., entropy, graph motifs) with advanced network analytics, including graph Laplacians and persistent homology. While their framework provides a mathematically rigorous anomaly indicator, it lacks model interpretability at the transaction level, which limits its applicability for forensic or regulatory tasks. Kumar et al. (2024) proposed an AI-blockchain framework that provides immutable data exchange and the prevention of numerous security threats, such as insider, Man-in-the-Middle (MitM), and impersonation attacks. The suggested blockchain mechanism guarantees secure communication between the involved entities of the smart healthcare system. By efficiently extracting temporal and spatial features using PSLSTM and MhSA mechanisms, the suggested deep learning model for detecting cyber threats can precisely identify attacks in smart healthcare systems. The proposed blockchain integration guaranteed the reliability of the data used to train the model. The suggested model for detecting cyber threats using deep learning is made more transparent by the SHAP-based interpretability mechanism. Similarly, Hasan et al. (2024) proposed using explainability analysis and machine learning classifiers to detect anomalies in Bitcoin transactions. They developed the XGBCLUS under-sampling algorithm and contrasted it with other under-sampling and oversampling methods. They used a dataset of 30,248,134 Bitcoin transactions, of which only 108 were classified as anomalous. The authors assessed the performance of both single and ensemble tree-based classifiers. In order to comprehend the model's decision-making process, the study emphasizes how key it is to address data imbalance and integrate explainable AI techniques.

An accountability and explainability framework for mobile robots was presented by Fernández-Becerra et al. (2024). The authors incorporated large language models to produce natural language justifications for the robot's actions as a way to increase openness and trust. Kumar et al. (2024) proposed a blockchain-enabled XAI framework for making cyber threat detection for smart healthcare systems. Blockchain was used to secure data exchange and a deep learning-based cyber threat detection model with parallel stacked long short-term memory and multi-head attention. The proposed model was evaluated on the ToN-IoT and IoT healthcare security datasets. The SHAP technique was used to interpret the model's decision-making process. Jayanthi et al. (2024) explored the integration of XAI and blockchain technology to enhance transparency, accountability, and trust in public administration. They examine how blockchain can effectively ensure secure and immutable record-keeping and XAI application in policy decisions, resource allocation, and service delivery. Ashfaq et al. (2022) propose a predictive model combining XGBoost and random forest algorithms with blockchain to detect fraudulent Bitcoin transactions, emphasizing smart contract integration for real-time classification. Likewise, Nadella et al. (2024) introduced a novel framework using K-means clustering to detect fraudulent activities without labeled data. Kilic et al. (2022) analyze blockchain transaction patterns to identify and flag suspicious activities using different machine learning algorithms with the aim to prevent fraud proactively.

Ranganatha and Mustafa (2025) proposed bidirectional 3D quasi-recurrent neural networks with blockchain to improve fraud detection in mobile transactions. Their proposed model outperforms existing models like KNN and decision tree by 12.09%, 8.91%, and 6.92% higher in accuracy. In Pranto et al. (2022), the paper proposes a machine learning and blockchain model for ensuring data privacy and fraud detection. Likewise, Osterrieder et al. (2024) provided a detailed examination of anomalies and frauds in blockchain networks. The paper concludes that spotting unusual activity and fraud within blockchain networks is vital for keeping these systems secure, trustworthy, and running smoothly (Osterrieder et al., 2024). Also, Elmougy and Liu (2023) presented a data science approach to detect fraudulent transactions and illicit nodes in the bitcoin network, introducing the Elliptic++ dataset. A predictive model that integrates graph-based representation learning and semantic feature extraction to detect complex fraud patterns in blockchain was discussed in Sheng et al. (2025). Their tests using massive, real-world blockchain data show that their new model is better than current methods in terms of accuracy, catching fraudulent activity, and avoiding missed cases. This emphasizes how significant it is to consider the connections between different parts of the blockchain and the meaning behind the transactions for strong fraud detection.

2.2 Explainable AI in Blockchain Systems

The lack of interpretability in ML models remains a challenge, especially in security-sensitive applications. Many studies have examined how explainable artificial intelligence (XAI) can address this challenge. For example, Rane et al. (2023) and Kumar et al. (2024) used SHAP values for interpreting the impact of each input feature to help stakeholders understand the model's decision. While several studies applied SHAP in cybersecurity or finance, only a few explored its potential in blockchain. However, explaining blockchain activity requires looking at how transactions move and how smart contracts operate, not just the individual data points. Other studies (Chahar et al., 2025; Hasan et al., 2024; Kovalenko, 2025; Pasupuleti, 2025) highlight the transformative synergy between AI and blockchain in finance. AI and blockchain create powerful possibilities together. For instance, they can enable secure digital identities, spot fraudulent activities, and even automatically verify financial records using smart contracts. While their research highlighted the importance of understanding transaction activity as it happens, they didn't provide specific methods or visual tools to actually achieve this understanding within blockchain systems.

Recent studies have highlighted the need for accurate blockchain fraud data. This data must truly show the complex ways decentralized transactions happen and be clearly labeled. However, many previous studies used synthetic data or small parts of blockchain activity (Sheng et al., 2025). These methods struggle to apply to real-world scams and manipulations. To fix this, we built a comprehensive dataset. We gathered real Ethereum transaction data using the Etherscan API. This data was then combined with scam address lists from projects like MyEtherWallet and Etherscan's community. Additionally, we added publicly available blockchain phishing datasets like Eth-PSD to increase the variety and coverage of our study data. Hence, our study data reflects authentic patterns of token transfers, transaction timing, contract usage, and wallet activity. This approach differs from simulation-based methods. We programmatically assigned labels by cross-referencing sender and receiver addresses with scam blacklists. Using these techniques enabled the identification of manipulated transactions without manual annotation. To further support robust modeling, we engineer behavioral and temporal features, which include inter-transaction time, address activity frequency, and transaction scheduling patterns. Unlike Hasan et al. (2024), we do not rely on under-sampling but instead integrate SMOTE and SHAP-based interpretation over enriched metadata. This multi-source, feature-rich dataset provides a realistic foundation for training and evaluating machine learning models under conditions that closely mirror operational blockchain environments. This method dramatically accelerates the detection of illicit transactions by making the computational process more efficient, as pointed out by Osterrieder et al. (2024).

Methodology

Figure 1 shows our proposed explainable AI framework for blockchain manipulation detection. We discuss details of each phase in the following section of this paper.

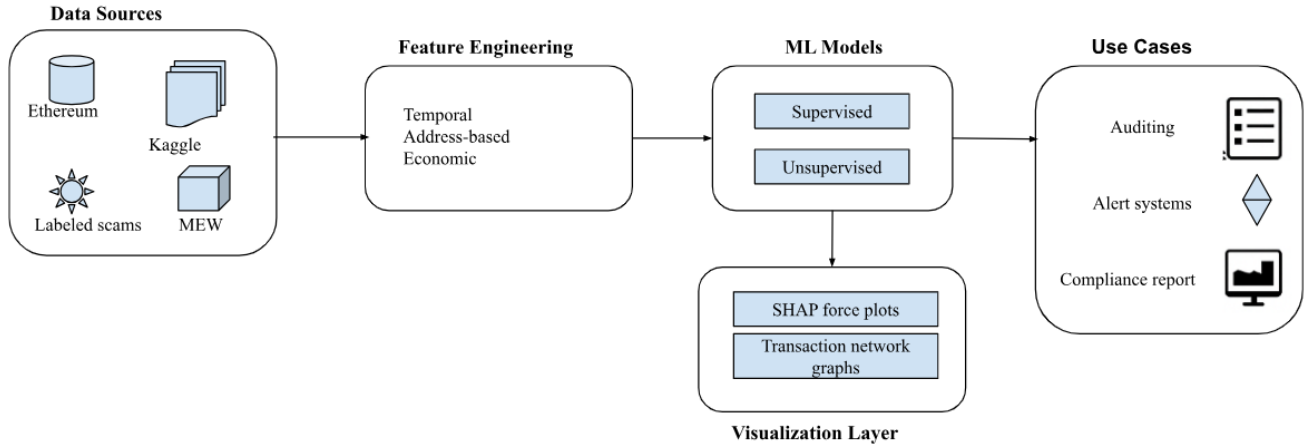


Figure 1: End-to-end explainable AI framework for blockchain transaction manipulation detection, illustrating multi-source data aggregation, feature engineering, supervised and unsupervised learning integration, and visualization-driven decision support.

3.1 Problem Formulation

We defined the blockchain transaction as set $G = (V, E)$, where V is the set of wallet addresses (nodes) and $E \subseteq V \times V$ represents transactions (edges) between them. Each edge $e_{ji} \in E$, is associated with a transaction tuple, as defined in Eq. (1).

where: t is the transaction timestamp, a_{ij} is the amount transferred, T_{ij} is the token type, γ_{ij} is the gas fee and ϕ_{ij} denotes interaction with smart contracts (binary). The goal is for the machine learning model to learn the function defined in Eq. (2). X is a feature space derived from historical transactions, and $f_{\theta}(x) = 1$ indicates a manipulated transaction. Additionally, we define an explanation function (see Eq. 3) that returns the contribution of each feature $x_j \in X$ to the prediction $f_{\theta}(x)$, using SHAP values (see Eq. 4) as defined in Kumar et al. (2024).

$$e_{ij} = (t, a_{ij}, T_{ij}, \gamma_{ij}, \phi_{ij}) \quad (1)$$

$$f_{\theta} : X \rightarrow \{0,1\} \quad (2)$$

$$\varepsilon(x, f_{\theta}) \rightarrow \mathbb{R}^d \quad (3)$$

$$\phi_j = E_{S \subset X \setminus \{x_j\}} [f(S \cup \{x_j\}) - f(S)] \quad (4)$$

3.2 Data Collection and Preprocessing

We constructed the study data by aggregating Ethereum-based transactions and tagging them using publicly known scam address lists and labeled phishing transaction datasets. Initially, ERC-20 token transfers, and general Ethereum transaction records were collected using the Etherscan API, capturing fields such as sender and receiver addresses, token metadata, and transaction costs. To obtain scam-related labels, we incorporated a curated blacklist of phishing-associated Ethereum addresses from the MyEtherWallet (MEW) Ethereum Lists (MyEtherWallet, n.d.). While additional attempts were made to scrape scam-labeled wallets from Etherscan's public tagging system, these were largely limited by dynamic page rendering and automation constraints. To augment the data, we integrated the Ethereum Phishing Scams Dataset (Eth-PSD) from Kaggle, which includes pre-labeled scam and non-scam transactions. Transactions were labeled as manipulated (label = 1) if the sender or receiver address matched an entry in the blacklist or if they originated from the Kaggle dataset. This procedure gave us a combined dataset of over 117,000 transactions (label 0 = 79,216, label 1 = 38,143). The Synthetic Minority Oversampling Technique (SMOTE) was applied during preprocessing to ensure sufficient representation of the minority class in the study data. The complete data preparation pipeline, including source collection, cleaning, merging, and labeling, is outlined in Algorithm 1. Table 1 summarizes the final feature set used in this study.

Algorithm 1: Blockchain Transaction Collection and Labeling

Input:

$A \leftarrow$ Etherscan API for Ethereum transactions
 $L \leftarrow$ Scam address list from MyEtherWallet (MEW)
 $K \leftarrow$ Ethereum Phishing Scams Dataset from Kaggle
 $T_{start}, T_{end} \leftarrow$ Time window for transaction scraping

Output:

$D_{labeled} \leftarrow$ Preprocessed and labeled transaction dataset

Procedure:

- 1: Initialize $D_{raw} \leftarrow \emptyset$
- 2: Retrieve transactions and token data from A within $[T_{start}, T_{end}]$
- 3: **for** each transaction $t_x \in$ API response, **do**
- 4: Extract: from, to, value, timestamp, contractAddress, input
- 5: Append t_x to D_{raw}
- 6: **end for**
- 7: Load $D_{kaggle} \leftarrow$ pre-labeled dataset K
- 8: Standardize schema of D_{raw} and D_{kaggle}
- 9: Combine: $D_{merged} \leftarrow D_{raw} \cup D_{kaggle}$
- 10: **for** each transaction $t_x \in D_{merged}$ **do**
- 11: **if** $t_x.from \in L$ or $t_x.to \in L$, **then**
- 12: $t_x.label \leftarrow 1$
- 13: **else if** t_x .label is undefined, **then**
- 14: $t_x.label \leftarrow 0$

```

15:   end if
16: end for

17: Handle missing values and clean formats
18: Return  $D_{labeled} \leftarrow$  cleaned and labeled dataset

```

Table 1 Dataset Feature Definitions and Descriptions

Feature	Description
contractaddress	Smart contract address involved in the transaction (if applicable).
from	Sender’s wallet address
to	Receiver’s wallet address
value	Token value transferred, measured in Wei (the smallest unit of ETH).
timestamp	Transaction timestamp (UNIX epoch time in seconds).
input	Encoded smart contract function call or transaction input data.
label	Transaction classification: 0 for normal, 1 for scam.

3.3 Feature Engineering

The “inter_tx_time” feature was computed by calculating the time difference (in seconds) between successive transactions from the same sender address. Table 1 shows a detailed overview of the features engineered in this study. We selected features based on their ability to capture behavioral, temporal, or economic patterns associated with manipulated blockchain activity. The temporal features (tx_hour, tx_day, and inter_tx_time) give insight into when and how frequently a wallet interacts with the blockchain. Additionally, the economic features (value and gas_to_value_ratio) mostly used in previous works help distinguish between high-value legitimate transfers and micro-manipulative behaviors. The address_tx_count feature helps measure the activity frequency for each sender address. This function makes it easy to identify automation-prone or bot-like behavior. These diverse features formed the foundational input space for our classification and anomaly detection models. All continuous features were normalized using Min-Max scaling. Missing values were dropped due to their insignificant effect on the total study data size, while categorical attributes were encoded. These features provided a foundational input space for supervised and semi-supervised models used in downstream detection tasks. This final feature set was informed by practical approaches outlined in recent works by Ikeda et al. (2025), Kumar et al. (2024), and Hasan et al. (2024). Algorithm 2 illustrates our feature engineering pipeline.

Algorithm 2 Feature Engineering and Preprocessing

Input: $D_{labeled} \leftarrow$ Labeled dataset with transaction metadata $F \leftarrow$ Set of selected features $S \leftarrow$ Oversampling strategy (SMOTE)**Output:** $D_{final} \leftarrow$ Feature-enhanced and class-balanced dataset**Procedure:**1: **for** each transaction $tx \in D_{labeled}$ **do**

2: Extract and compute:

3: $tx.tx_hour \leftarrow hour(tx.timestamp)$ 4: $tx.tx_day \leftarrow day_of_week(tx.timestamp)$ 5: $tx.inter_tx_time \leftarrow$ time since previous tx from same address6: $tx.address_tx_count \leftarrow$ frequency of tx.from in $D_{labeled}$ 7: **end for**8: Select feature set $F \leftarrow \{value, tx_hour, tx_day, inter_tx_time, address_tx_count\}$ 9: **if** class imbalance is detected, **then**10: Apply oversampling using S to generate $D_{balanced}$ 11: **Return** $D_{final} \leftarrow D_{balanced}$ 12: **else**13: **Return** $D_{final} \leftarrow D_{labeled}$ with F 14: **end if**

Table 2 Summary of Engineered Features Used in Model Training and Evaluation

Feature Name	Description	Feature Category	Data Type
value	Transaction amount measured in Ether (ETH) or equivalent token units.	Economic	Numerical
tx_hour	Hour of the day when the transaction was initiated (0–23).	Temporal	Categorical
tx_day	Day of the week when the transaction was initiated (0 = Monday, 6 = Sunday).	Temporal	Categorical
inter_tx_time	Time interval (in seconds) between successive transactions from the same sender.	Temporal	Numerical
address_tx_count	Total number of transactions initiated by the sender address.	Address Activity Metric	Numerical
gas_to_value_ratio	Ratio of gas consumed to value transferred during a transaction (evaluated in earlier feature sets).	Economic	Numerical

3.4 Learning Models and Training Configuration

3.4.1 Model Architecture

We implemented supervised and unsupervised machine learning models to detect fraudulent blockchain transactions. We adopted the XGBoost and LightGBM as classification models to detect manipulated blockchain transactions, due to their robustness on structured or tabular data (Ashfaq et al., 2022). Additionally, these models can effectively handle non-linear relationships and feature interactions, which are common in transaction behavior patterns. For unsupervised anomaly detection, we used:

- Autoencoder neural networks (Eq. 6), designed with a symmetric structure (*input* $\rightarrow$ *bottleneck* $\rightarrow$ *reconstruction*), consisting of three fully connected layers with ReLU activation for the encoder and decoder and a bottleneck of 32 units. Figure 2 shows the architecture of our autoencoder.
- Isolation Forest, which recursively partitions the data using randomly selected features and thresholds, isolating anomalies as those with short average path lengths.

The final output of each model is a binary label indicating normal (0) or manipulated (1) transactions. We defined the loss function as minimized:

$$L_{binary} = \frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (5)$$

$$L_{AE} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2 \quad (6)$$

where $\hat{x}_i$ is the reconstructed input.

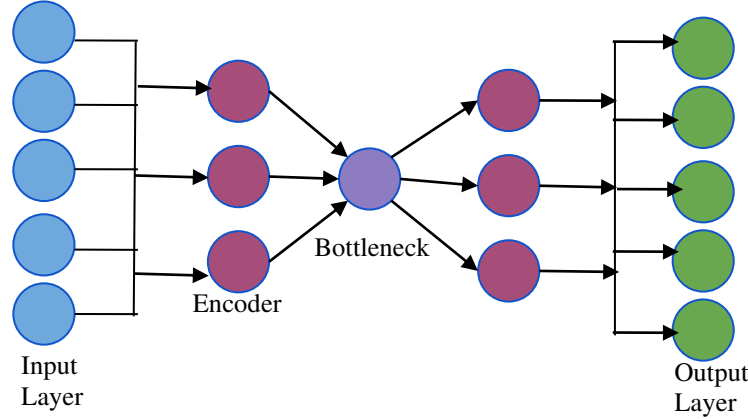


Figure 2: Simplified architecture of the deep autoencoder model used for blockchain transaction anomaly detection, highlighting the encoder, bottleneck, and decoder layers.

We benchmarked our models with (1) logistic regression with L2 regularization, (2) random forest with 100 trees, and (3) one-class support vector machine (SVM) for anomaly detection using the RBF kernel. These models were selected to reflect commonly used baselines in blockchain anomaly detection and financial fraud literature.

3.4.2 Training Configuration

We split our cleaned data into an 80% train set and a 20% test set to train and test all models. The stratified 5-fold cross-validation was adopted to ensure robustness across class distributions. Our deep learning autoencoder hyperparameters were: Adam optimizer, a learning rate of 0.001, binary cross-entropy loss, a batch size of 128, and epochs of 50 with early stopping (patience = 5) to prevent overfitting. For the XGBoost model, we set the following hyperparameters after tuning: max_depth = 6, n_estimators = 200, learning_rate = 0.1, subsample = 0.8. For LightGBM: num_leaves = 31, max_depth = -1, learning_rate = 0.05, n_estimators = 250. The autoencoder used dropout (rate = 0.3) between dense layers and employed min-max scaled inputs. Reconstruction error was used to compute anomaly scores, with a threshold set using the 95th percentile on benign validation samples. The final features used to train our models were “value,” “tx_hour,” “tx_day,” “inter_tx_time,” and “address_tx_count.” A random seed value of 42 was set for all models to ensure reproducibility.

3.4.3 Hyperparameter Optimization And Model Evaluation

The random search optimization algorithm was adopted in this study to tune the hyperparameters of our tree-based models. We set the iteration to 20 and evaluated using cross-validated F1-score. For the autoencoder, we evaluated bottleneck sizes (16, 32, 64), dropout rates (0.1--0.5), and learning rates (0.001--0.01). The best-performing configuration was retained based on validation reconstruction error and ROC-AUC. We evaluated our model's performance using precision, recall, F1-score, and area under the ROC curve (AUC) metrics. Definitions are given in equations 7 and 8 below. The heterogeneous combination of evaluation metrics is suitable for imbalanced binary classification.

$$Precision = \frac{TP}{TP + FP}, Recall = \frac{TP}{TP + FN}, F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (7)$$

$$AUC = \int_0^1 TPR(FPR) dFPR \quad (8)$$

where TP = True Positives (correctly predicted positives), TN = True Negatives (correctly predicted negatives), FP = False Positives (incorrectly predicted positives), FN = False Negatives (incorrectly predicted negatives), and TPR (FPR) = True Positive Rate as a function of the False Positive Rate.

3.5 Model Explainability

We employed the SHapley Additive exPlanations (SHAP) for the model's interpretability. This approach enables the transformation of model predictions into feature-level contributions to offer two complementary views (local and global) of interpretability. At a global level, SHAP provides an aggregated measure of feature importance across the entire dataset. By this method we are able to identify which variables consistently influence the model's outputs. Locally, SHAP offers instance-level explanations by quantifying how individual features drive specific predictions. This observation is useful for investigating manipulated transactions flagged by the model. Additionally, we utilized SHAP visualization tools, including summary plots and force plots, to further improve interpretability and stakeholder auditability. This interpretability of machine learning promotes transparency and trustworthiness in fraud detection. Additionally, it follows best practices for AI in finance and cybersecurity, as highlighted in recent work (Kumar et al., 2024; Rane et al., 2023) on deploying explainable AI in sensitive areas.

4. Results and Discussions

This section presents the outcome of our proposed multi-source explainable AI for detecting manipulated blockchain transactions to enhance trust in decentralized systems.

4.1 Models Performance

Figure 3 shows the performance plot of the XGBoost and LightGBM models. As seen from Figure 4, both XGBoost and LightGBM achieved exceptional classification performance. The XGBoost obtained an F1 score of 0.9685, while LightGBM followed closely with 0.9681. Additionally, we observed that these high scores were supported by precision and recall values exceeding 0.965 for both models (see Figure 3). These results show their ability to accurately identify manipulated transactions with minimal false positives and false negatives. From the confusion matrix plot (see Fig. 4), the XGBoost classifier correctly classified 15,392 scam transactions and 15,294 normal transactions, with only 451 false negatives and 550 false positives (see Figure 4). The LightGBM classifier obtained similar results but misclassified slightly fewer normal transactions (543) and somewhat more scam instances (470). In contrast, the unsupervised isolation forest and autocondensor neural network models perform poorly, with high misclassification rates in both classes. We observed an F1-score of 0.26, with a precision of 0.49 and a recall of only 0.18 for the isolation forest. The confusion matrix (see Figure 4) shows that over 12,900 scam transactions were misclassified as normal by the isolation forest model, resulting in a high false negative rate. Thus, without label guidance, the isolation forest fails to capture the nuances necessary for effective manipulation detection. These observations highlight the value of using labeled features and gradient-boosted methods for fraud detection in blockchain environments. These findings illustrate the limitations of applying unsupervised anomaly detection in complex domains such as blockchain fraud, where feature patterns may be subtle and entangled with legitimate behavior.

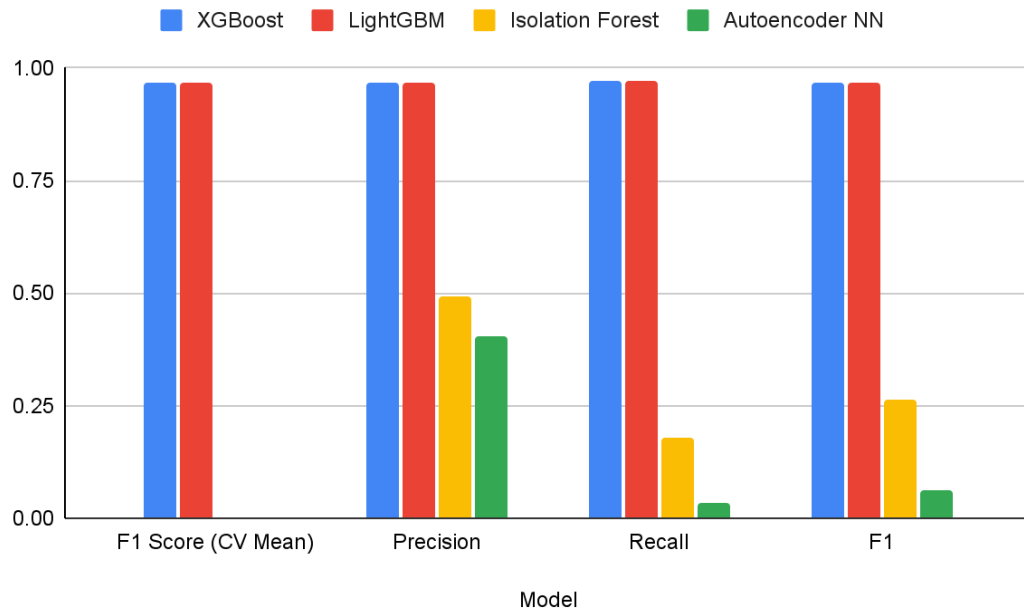


Figure 3 Comparative performance metrics for anomaly detection models (XGBoost, LightGBM, Isolation Forest, and Autoencoder neural network) based on F1 score (cross-validation mean), precision, recall, and F1 score on the test dataset.

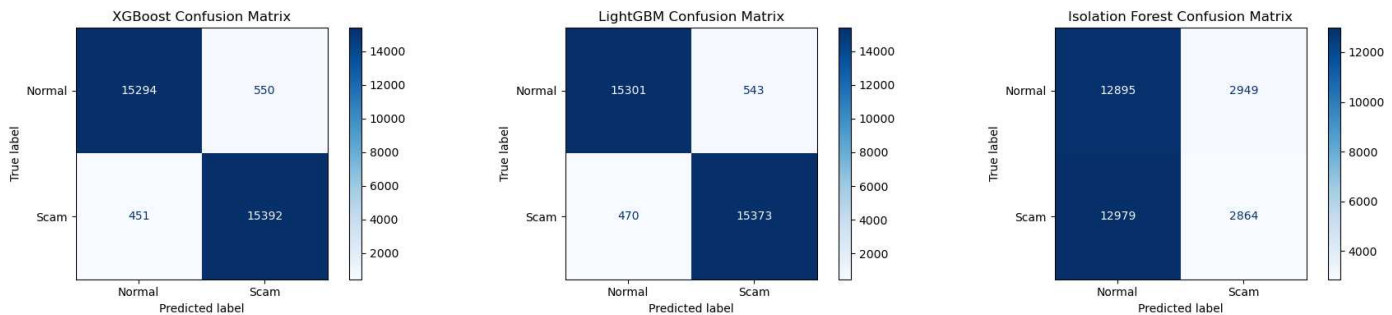


Figure 4 Confusion matrices of XGBoost, LightGBM, and Isolation Forest models for blockchain transaction anomaly detection, illustrating the classification of normal versus scam transactions on the test dataset.

Figure 5 shows the Receiver Operating Characteristic (ROC) curves for the XGBoost, LightGBM, and isolation forest models. Again, the XGBoost and LightGBM achieved an area under the curve score of 1.00. This result indicates perfect separation between fraudulent and legitimate transactions. The results demonstrate their suitability for tasks involving detecting fraud in tabular data using behavioral blockchain features. Conversely, the isolation forest model performed no better than random guessing ($AUC = 0.50$), which is equivalent to random guessing. This result further confirms the limitations of purely unsupervised detection in this domain. These results affirm the suitability of gradient-boosted decision trees for blockchain anomaly detection tasks.

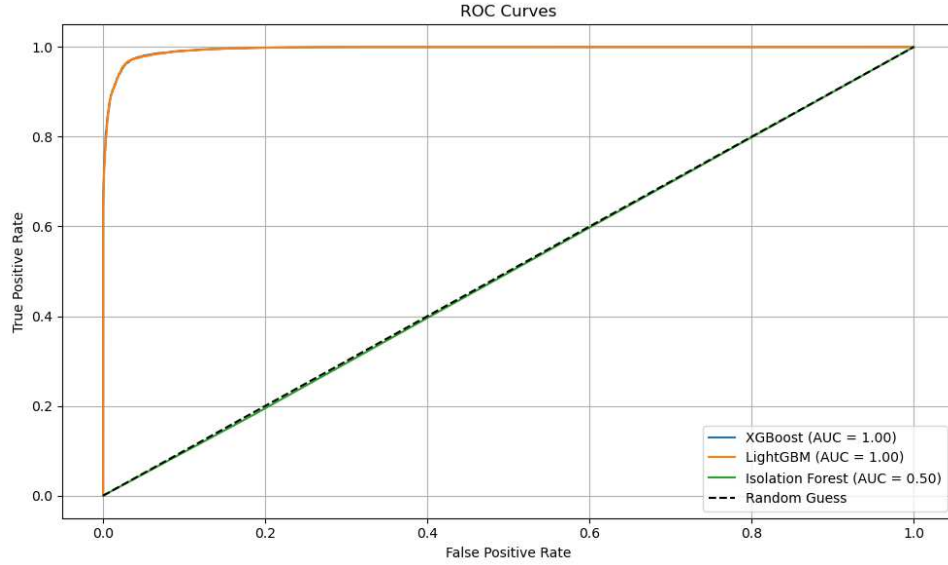


Figure 5 Receiver operating characteristic (ROC) curves of XGBoost, LightGBM, and Isolation Forest models for blockchain transaction anomaly detection, with AUC scores indicating classification performance on the test dataset.

Figure 6 shows the autoencoder training and validation loss and reconstruction error distribution plots. It is worth noting that the deep autoencoder neural network was trained exclusively on benign (label = 0) transactions. The early stopping mechanism halted training at epoch 6 due to convergence in validation loss. It was observed that both loss curves demonstrate stability with no signs of overfitting or divergence, indicating that the model adequately learned to reconstruct normal transaction patterns. We also observe a clear separation between normal and anomalous transactions near the lower tail in the reconstruction error distribution. We empirically selected the anomaly detection threshold at the 95th percentile of reconstruction errors on benign validation data, resulting in a cutoff of 0.00037. This threshold was then applied to classify test transactions: those exceeding the threshold were flagged as anomalous (label = 1). Despite its ability to learn the benign distribution, the autoencoder achieved moderate classification results when used for binary anomaly detection (see Fig. 7). Its performance was lower than supervised models, which is consistent with expectations given the lack of fraud supervision. Nevertheless, it still contributes to an ensemble approach or as a component in pre-screening layers for real-time fraud detection systems.

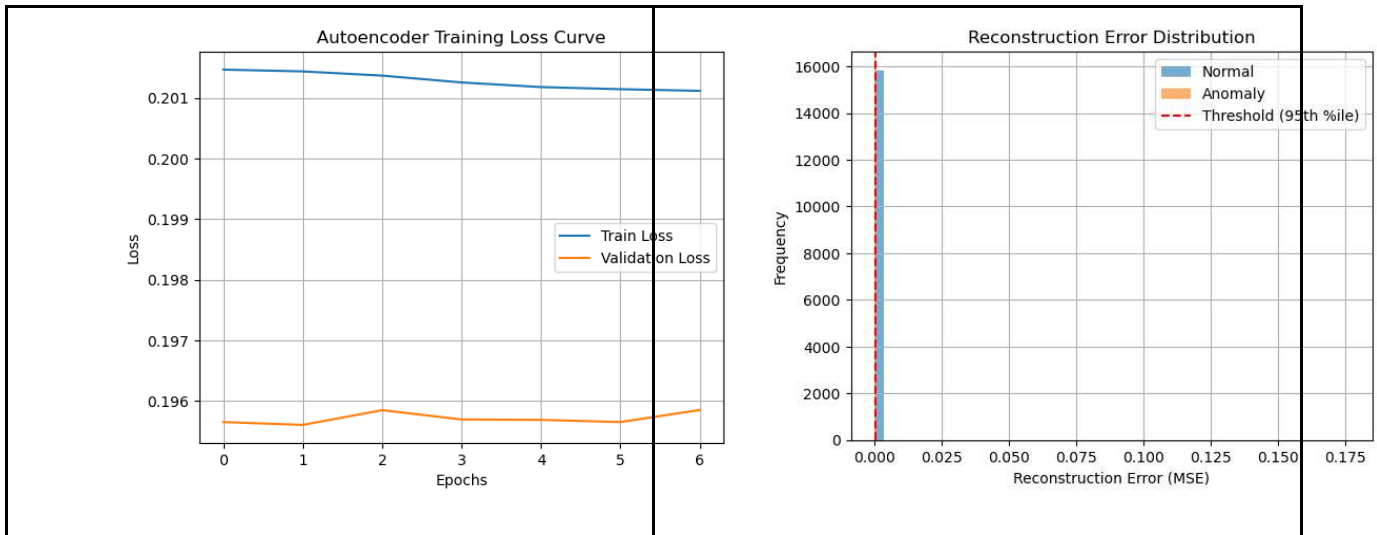


Figure 6 (Left) Autoencoder training and validation loss curves showing convergence behavior over epochs. (Right) Reconstruction error distribution for blockchain transaction anomaly detection, with the 95th percentile threshold used to separate normal and anomalous transactions.

Figure 7 shows the confusion matrix and ROC curve plots of the autoencoder. It was observed that the autoencoder model obtained a relatively high true negative rate (15,051 correctly identified normal transactions) but misclassified the vast majority of scam transactions (15,302 false negatives), detecting only 541 true positives. This condition leads to a high false negative rate, which is a key concern in fraud detection systems. The ROC curve confirms this observation. The model's AUC of 0.49 suggests the model performs no better than random guessing. This is true even when we apply a reconstruction threshold based on the 95th percentile of the benign reconstruction error distribution. These findings point out the inherent limitations of purely unsupervised models such as autoencoders and one-class SVMs when applied in domains with subtle behavioral differences between classes and high intra-class variability. Thus, without access to explicit fraud signals during training, the model struggles to generalize to the structural and economic patterns characteristic of scam behavior.

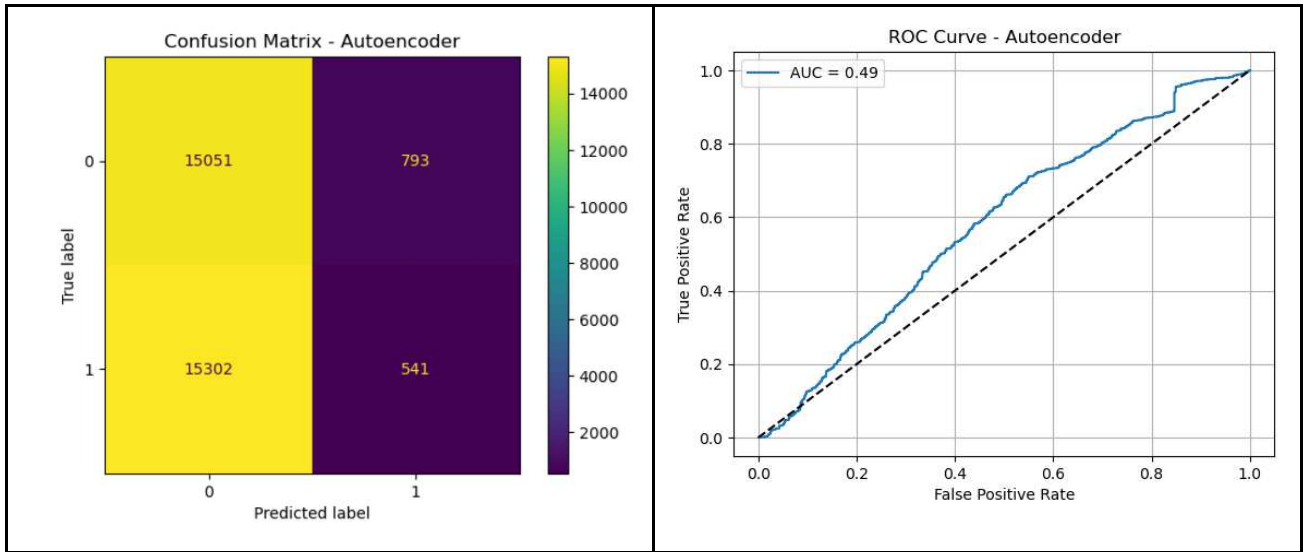


Figure 7 (Left) Confusion matrix showing classification results of the autoencoder model for blockchain transaction anomaly detection. (Right) Receiver operating characteristic (ROC) curve of the autoencoder model, with an area under the curve (AUC) score of 0.49.

4.2 Mode Explainability and Feature Interpretability with SHAP

We applied SHapley Additive exPlanations (SHAP) analysis to both XGBoost and LightGBM models. The summary plots illustrated the distribution and magnitude of feature impacts across the dataset. While force plots are enabled, it helps understand how features contribute positively or negatively to a given classification outcome. Figures 8 and 9 show the SHAP summary plots for the LightGBM and XGBoost models, respectively. For both models, we observed that the feature “address_tx_coun” was the most influential. This observation suggests that higher transaction volumes from a single wallet (colored red on the right) strongly increased the likelihood of being flagged as manipulated. Our observations align well with behavioral patterns commonly observed in automated or bot-driven scam accounts that execute bursts of activity. The second most important feature observed to contribute significantly to both models’ classification decisions was “value.” The evidence suggests that lower-value transfers (blue points on the left) are generally associated with non-manipulated transactions, while higher values give an ambiguous impact. This discovery reveals that value alone is insufficient without contextual behavior. Also, we saw that “inter_tx_time,” which captures the time elapsed between successive transactions by the same address, was the third most influential feature. On the other hand, we observed that short inter-transaction times had a negative SHAP value. This observation suggests that they are capable of signaling suspicious patterns such as flash loan attacks or transaction loops. Additionally, it was observed that temporal features such as “tx_hour” and “tx_day” had lower but still non-negligible contributions to the model’s decision-making. The link observed between scam incidents and specific temporal patterns, such as off-peak hours or specific days of the week, provides an intriguing insight. This result is consistent with research on the behavior of digital fraud. The finding highlights common tactics employed by attackers across different online platforms and services. The consistent patterns revealed in the SHAP plots for both XGBoost and LightGBM increase confidence in the robustness of the features engineered within this study. Also, it affirms the necessity of explainable AI applications in the complex domain of blockchain fraud detection. These results show that carefully constructed, interpretable supervised models hold considerable promise for deployment within DeFi ecosystems and non-fungible token marketplaces. Their capacity to flag potentially illicit transactions in real time could

substantially alleviate the workload of human analysts, offering crucial support for the development of sophisticated, automated blockchain forensic pipelines.

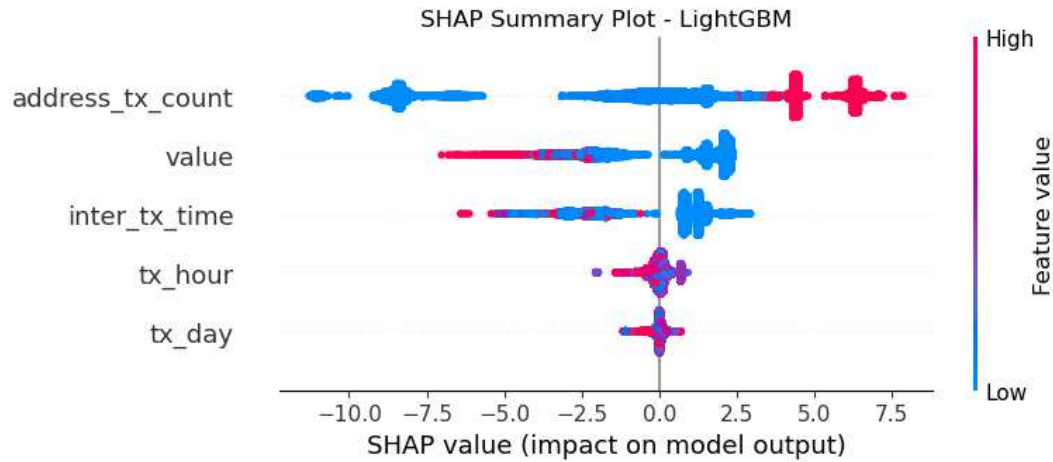


Figure 8 SHAP summary plot for the LightGBM model, illustrating feature importance and the impact of feature values on blockchain transaction anomaly detection.

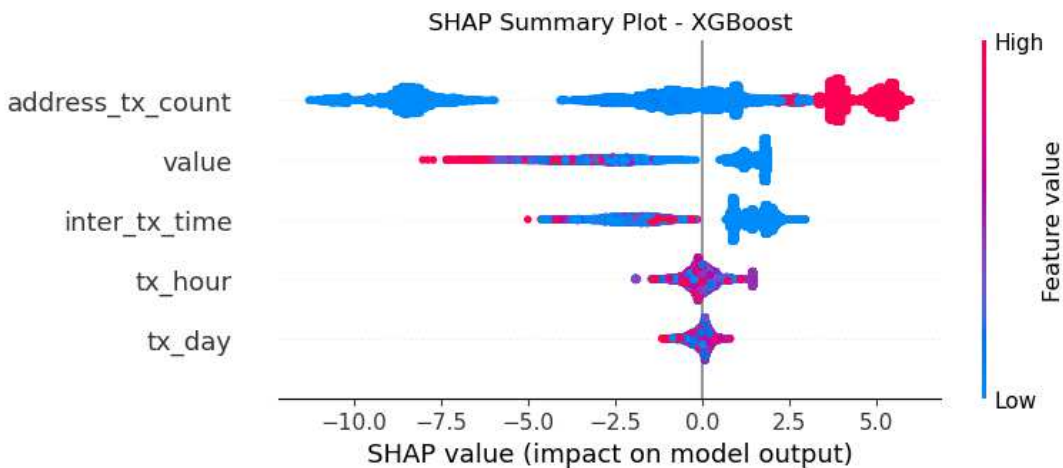


Figure 9 SHAP summary plot for the XGBoost model, highlighting the influence of address activity, transaction value, and temporal features on classification outcomes in blockchain transaction anomaly detection.

4.3 Benchmark Model Evaluation

We compared how well our models performed against three popular models (logistic regression with L2 regularization, random forest with 100 trees, and one-class support vector machine (SVM) using an RBF kernel) in this area to confirm that our classifiers are effective. Table 3 shows the results of these benchmark models. Among the baselines, random forest was seen to demonstrate the strongest performance, achieving an F1 score of 0.9722, precision of 0.9677, and an AUC of 0.996. These performances closely match those of the XGBoost and LightGBM. This observation confirms the strength of ensemble tree-based methods in capturing non-linear patterns in structured transaction data. However, logistic regression, though it achieved high precision (0.9838), suffered from a low recall (0.6001), resulting in a modest F1-score of 0.7455. This result indicates that although the model correctly classified many scam transactions it flagged, it failed to identify a large portion of actual scams. This might be due to its linear decision boundaries being insufficient for the complex structure of blockchain behavior. On the other hand, the one-class SVM trained as an unsupervised classifier achieved better performance than the isolation forest. However, it underperformed with an F1 score of 0.7409 and an AUC of 0.7843 when compared with supervised models. As seen, the SVM maintained high precision (0.9277), but its recall (0.6167) highlights the challenge of detecting a wide range of

manipulation patterns without labeled data. These observations reinforce that supervised, non-linear models are the best fit for fraud detection tasks within blockchain environments compared with linear and unsupervised models.

Table 3: Performance Metrics of Benchmark Models for Blockchain Transaction Anomaly Detection

Model	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	0.98	0.60	0.75	0.84
Random Forest	0.97	0.98	0.97	1.00
One-Class SVM	0.93	0.62	0.74	0.78

5. Conclusion and Future Work

This study presents a multi-source explainable machine learning framework for detecting manipulated blockchain transactions, specifically within Ethereum-based networks. Our approach integrates engineered behavioral, temporal, and structural features derived from transaction metadata and wallet activity. We evaluated four models, namely XGBoost, LightGBM, isolation forest, and autoencoder neural network on a balanced and feature-enriched dataset aggregated from Ethereum transaction logs and known scam addresses. We observed that the supervised classification models obtained a higher blockchain fraud detection performance. XGBoost obtained an F1 score of 0.9685, and LightGBM achieved 0.9681, demonstrating consistent effectiveness across 5-fold stratified cross-validation. Both models balanced precision (0.965–0.966) and recall (0.970–0.971), highlighting their reliability in detecting fraudulent transactions without overfitting. In contrast, the unsupervised isolation forest approach showed limited effectiveness, with an F1 score of 0.26 and recall of 18.1%. Likewise, the autoencoder neural network confirms the benefit of using labeled behavioral features in this domain. Additionally, we employed SHAP explainability methods to interpret model behavior. It was seen that “address_tx_count,” “value,” and “inter_tx_time” were the most significant features across both XGBoost and LightGBM. These findings suggest that high-frequency transaction activity and short inter-transaction intervals are reliable indicators of suspicious behavior in decentralized financial systems.

Although our results demonstrate strong performance in a controlled and labeled setting, there are a few limitations. First, the reliance on static scam lists (e.g., from MyEtherWallet and Etherscan tags) may exclude emerging and novel manipulation techniques. Second, the method did not cover cross-contract interactions or multi-chain activity; it was restricted to ERC-20 token transactions. Future research will extend this framework in the following directions:

- Incorporate transaction behavior across multiple blockchains, for example, BNB Chain, and Polygon. We will use interoperable APIs and big data pipelines to identify manipulation at a broader ecosystem level.
- Integrate this study into a real-time streaming architecture for proactive blockchain fraud mitigation.
- Investigate the impact of adversarial attacks (e.g., camouflage transactions or adversarial wallets) on detection accuracy. Also apply defense techniques to improve model resilience.
- Identify latent scam clusters without full reliance on labeled data by exploring semi-supervised and graph-based learning models.
- Build an interface that combines SHAP outputs with visual transaction flows to support forensic analysts, smart contract auditors, and compliance officers in understanding flagged anomalies.

In conclusion, this study provides a robust and understandable solution for automated blockchain forensics and lays the groundwork for trustworthy fraud detection in decentralized environments.

Declarations

Availability of data and materials: All datasets used in this paper are publicly available and can also be made available upon a reasonable request.

Competing interests: The authors declare that they have no competing interests

Funding: Not applicable

Authors' contributions: I.K.N. designed the study's methodology, conducted the experiments, and drafted the initial manuscript. O. N-B. and S. B. contributed to data analysis and manuscript revision. B.A.W. and A.F.A. supervised the project and reviewed the manuscript for accuracy and clarity. All authors reviewed the manuscript.

List of Abbreviations

AI - Artificial Intelligence
XAI - Explainable Artificial Intelligence
ML - Machine Learning
SHAP - SHapley Additive exPlanations
XGBoost - eXtreme Gradient Boosting
LightGBM - Light Gradient Boosting Machine
DeFi - Decentralized Finance
NFT - Non-Fungible Token
SMOTE - Synthetic Minority Over-sampling Technique
API - Application Programming Interface
ETH - Ethereum
MEW - MyEtherWallet
Eth-PSD - Ethereum Phishing Scams Dataset
ROC - Receiver Operating Characteristic
AUC - Area Under the Curve
RBF - Radial Basis Function
SVM - Support Vector Machine

References

- Ashfaq, T., Khalid, R., Yahaya, A. S., Aslam, S., Azar, A. T., Alsafari, S., & Hameed, I. A. (2022). A machine learning and blockchain based efficient fraud detection mechanism. *Sensors*, 22(19), 7162. <https://doi.org/10.3390/s22197162>
- Bello, H. O., Idemudia, C., & Iyelolu, T. V. (2024). Integrating machine learning and blockchain: Conceptual frameworks for real-time fraud detection and prevention. *World Journal of Advanced Research and Reviews*, 23(1), 056–068. <https://doi.org/10.30574/wjarr.2024.23.1.1985>
- Chahar, S., Kaur, K., Kaswan, K. S., & Dhattewal, J. S. (2025). Explainable AI in blockchain system for decentralized governance. *2025 International Conference on Pervasive Computational Technologies (ICPCT)*, 725–729. <https://doi.org/10.1109/icpct64145.2025.10941600>
- Chainalysis Team. (2025, January 15). *2025 crypto crime trends from chainalysis*. Chainalysis. <https://www.chainalysis.com/blog/2025-crypto-crime-report-introduction>
- Chitsungo, C. (2024). Harnessing digital strategies to combat cryptocurrency-enabled crimes: Addressing money laundering, illicit trade, and cyber threats. *American Journal of International Relations*, 9(7), 77–106. <https://doi.org/10.47672/ajir.2523>
- Chowdhury, N., Deka, G. C., & Sufian, M. A. (2024). Developing trading strategies in decentralized market prediction by using AI, ML, and blockchain technology. In *Blockchain and AI* (pp. 58–122). CRC Press. <https://doi.org/10.1201/9781003162018-3>
- Elmougy, Y., & Liu, L. (2023). Demystifying fraudulent transactions and illicit nodes in the bitcoin network for financial forensics. *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 3979–3990. <https://doi.org/10.1145/3580305.3599803>
- Etherscan. (n.d.). *Ethereum (ETH) blockchain explorer*. Ethereum (ETH) Blockchain Explorer. Retrieved April 16, 2025, from <http://etherscan.io>
- Fernández-Becerra, L., González-Santamarta, M. Á., Guerrero-Higuera, Á. M., Rodríguez-Lera, F. J., & Matellán-Olivera, V. (2024). *Enhancing trust in autonomous agents: An architecture for accountability and explainability through blockchain and large language models*. Elsevier BV. <https://doi.org/10.2139/ssrn.4773675>
- Hasan, M., Rahman, M. S., Janicke, H., & Sarker, I. H. (2024). Detecting anomalies in blockchain transactions using machine learning classifiers and explainability analysis. *Blockchain: Research and Applications*, 5(3), 100207. <https://doi.org/10.1016/j.bcra.2024.100207>
- Hernandez Aros, L., Bustamante Molano, L. X., Gutierrez-Portela, F., Moreno Hernandez, J. J., & Rodríguez Barrero, M. S. (2024). Financial fraud detection through the application of machine learning techniques: A literature review. *Humanities and Social Sciences Communications*, 11(1). <https://doi.org/10.1057/s41599-024-03606-0>
- Ikeda, Y., Hadfi, R., Ito, T., & Fujihara, A. (2025). Anomaly detection and facilitation AI to empower decentralized autonomous organizations for secure crypto-asset transactions. *AI & SOCIETY*. <https://doi.org/10.1007/s00146-024-02166-w>

- Jayanthi, S., Suresh Kumar, N., Balashivudu, U., Purushotham, M., & Jeeva, S. (2024). An explorative study of explainable AI and blockchain integration in public administration. In *Applications of Blockchain and Artificial Intelligence in Finance and Governance* (pp. 230–271). CRC Press. <https://doi.org/10.1201/9781003518365-10>
- Kayikci, S., & Khoshgoftaar, T. M. (2024). Blockchain meets machine learning: A survey. *Journal of Big Data*, 11(1). <https://doi.org/10.1186/s40537-023-00852-y>
- Kilic, B., Sen, A., & Ozturan, C. (2022). Fraud Detection in Blockchains using Machine Learning. *2022 Fourth International Conference on Blockchain Computing and Applications (BCCA)*, 214–218. <https://doi.org/10.1109/bcca55292.2022.9922045>
- Kovalenko, A. (2025). *Sociotechnical governance of ai-driven credit systems: Integrating fairness metrics, blockchain transparency, and stakeholder dynamics in traditional and decentralized finance*. Institute of Electrical and Electronics Engineers (IEEE). <https://doi.org/10.36227/techrxiv.174285066.67574270/v1>
- Kshetri, N. (2025). Building trust in AI: How blockchain enhances data integrity, security, and privacy. *Computer*, 58(2), 63–70. <https://doi.org/10.1109/mc.2024.3505012>
- Kumar, P., Javeed, D., Kumar, R., & Islam, A. K. M. N. (2024). Blockchain and explainable AI for enhanced decision making in cyber threat detection. *Software: Practice and Experience*, 54(8), 1337–1360. <https://doi.org/10.1002/spe.3319>
- Mattackal, L. (2025, February 14). Crypto scams likely set new record in 2024 helped by AI, Chainalysis says. *Reuters*. https://www.reuters.com/technology/crypto-scams-likely-set-new-record-2024-helped-by-ai-chainalysis-says-2025-02-14/?utm_source=chatgpt.com
- Mondoh, B. S. (2025, February 26). Decoding Crypto Crime: Insights from TRM Labs' 2025 crypto crime report. *The Barrister Group*. https://thebarristergroup.co.uk/blog/insights-from-trm-labs-2025-crypto-crime-report?utm_source=chatgpt.com
- Mounnan, O., Manad, O., Boubchir, L., El Mouatasim, A., & Daachi, B. (2024). A review on deep anomaly detection in blockchain. *Blockchain: Research and Applications*, 5(4), 100227. <https://doi.org/10.1016/j.bcr.2024.100227>
- MyEtherWallet. (n.d.). *GitHub - MyEtherWallet/ethereum-lists: A repository for maintaining lists of things like malicious URLs, fake token addresses, and so forth. We love lists*. GitHub. Retrieved April 17, 2025, from <https://github.com/MyEtherWallet/ethereum-lists>
- Nadella, G. S., Meduri, K., Gonaygunta, H., Satish, S., & Pillai, S. E. V. S. (2024). Blockchain fraud detection using unsupervised learning: Anomalous transaction patterns detection using k-means clustering. *Proceedings of the 2024 Sixteenth International Conference on Contemporary Computing*, 407–412. <https://doi.org/10.1145/3675888.3676080>
- Osterrieder, J., Chan, S., Chu, J., Zhang, Y., Misheva, B. H., & Mare, C. (2024, February 17). *Enhancing security in blockchain networks: Anomalies, frauds, and advanced detection techniques*. arXiv.Org. <https://arxiv.org/abs/2402.11231>
- Pasupuleti, M. K. (2025). *Automated smart contracts: AI-powered blockchain technologies for secure and intelligent decentralized governance*. National Education Services. <https://doi.org/10.62311/nexs/rrv425>
- Pranto, T. H., Hasib, K. T. A. Md., Rahman, T., Haque, A. B., Islam, A. K. M. N., & Rahman, R. M. (2022). Blockchain and machine learning for fraud detection: A privacy-preserving and adaptive incentive based approach. *IEEE Access*, 10, 87115–87134. <https://doi.org/10.1109/access.2022.3198956>
- Rane, N., Choudhary, S., & Rane, J. (2023). Blockchain and Artificial Intelligence (AI) integration for revolutionizing security and transparency in finance. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4644253>
- Ranganatha, H. R., & Mustafa, A. S. (2025). Enhancing fraud detection efficiency in mobile transactions through the integration of bidirectional 3d Quasi-Recurrent Neural network and blockchain technologies. *Expert Systems with Applications*, 260, 125179. <https://doi.org/10.1016/j.eswa.2024.125179>
- Sheng, Z., Song, L., & Wang, Y. (2025, January 3). *Dynamic feature fusion: Combining global graph structures and local semantics for blockchain fraud detection*. arXiv.Org. <https://arxiv.org/abs/2501.02032>
- Singh, M. (2024, December 19). Losses from crypto hacks jump to \$2.2 bln in 2024, report says. *Reuters*. <https://www.reuters.com/technology/losses-crypto-hacks-jump-22-bln-2024-report-says-2024-12-19>
- Svetinovic, D. (2019). Keynote 3: Blockchain — iot — AI trust nexus. *2019 Sixth International Conference on Social Networks Analysis, Management and Security (SNAMS)*, 3–3. <https://doi.org/10.1109/snams.2019.8931835>
- Tabaghsar, B. J., Tavoli, R., & Toosi, M. M. A. (2024). Detecting anomalies in Bitcoin blockchain transactions using machine learning approach and extracting effective features based on Relief algorithm. *2024 10th International Conference on Artificial Intelligence and Robotics (QICAR)*, 286–295. <https://doi.org/10.1109/qicar61538.2024.10496617>
- Vajjhala, N. R., Fonkam, M., & Eappen, P. (2025). Introduction to explainable AI (XAI) and blockchain. In *Explainable AI and Blockchain for Secure and Agile Supply Chains* (pp. 1–15). Chapman and Hall/CRC. <https://doi.org/10.1201/9781003497363-1>